

# P2N2222A

## Amplifier Transistors

### NPN Silicon

#### Features

- These are Pb-Free Devices\*

#### MAXIMUM RATINGS ( $T_A = 25^\circ\text{C}$ unless otherwise noted)

| Characteristic                                                                         | Symbol         | Value          | Unit                       |
|----------------------------------------------------------------------------------------|----------------|----------------|----------------------------|
| Collector – Emitter Voltage                                                            | $V_{CEO}$      | 40             | Vdc                        |
| Collector – Base Voltage                                                               | $V_{CBO}$      | 75             | Vdc                        |
| Emitter – Base Voltage                                                                 | $V_{EBO}$      | 6.0            | Vdc                        |
| Collector Current – Continuous                                                         | $I_C$          | 600            | mA dc                      |
| Total Device Dissipation @ $T_A = 25^\circ\text{C}$<br>Derate above $25^\circ\text{C}$ | $P_D$          | 625<br>5.0     | mW<br>mW/ $^\circ\text{C}$ |
| Total Device Dissipation @ $T_C = 25^\circ\text{C}$<br>Derate above $25^\circ\text{C}$ | $P_D$          | 1.5<br>12      | W<br>mW/ $^\circ\text{C}$  |
| Operating and Storage Junction<br>Temperature Range                                    | $T_J, T_{stg}$ | -55 to<br>+150 | $^\circ\text{C}$           |

#### THERMAL CHARACTERISTICS

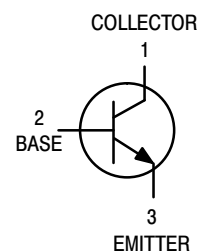
| Characteristic                          | Symbol          | Max  | Unit               |
|-----------------------------------------|-----------------|------|--------------------|
| Thermal Resistance, Junction to Ambient | $R_{\theta JA}$ | 200  | $^\circ\text{C/W}$ |
| Thermal Resistance, Junction to Case    | $R_{\theta JC}$ | 83.3 | $^\circ\text{C/W}$ |

Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

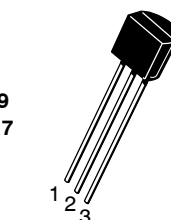


ON Semiconductor®

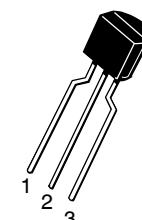
<http://onsemi.com>



TO-92  
CASE 29  
STYLE 17

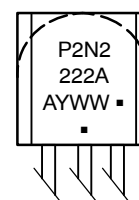


STRAIGHT LEAD  
BULK PACK



BENT LEAD  
TAPE & REEL  
AMMO PACK

#### MARKING DIAGRAM



A = Assembly Location

Y = Year

WW = Work Week

▪ = Pb-Free Package

(Note: Microdot may be in either location)

#### ORDERING INFORMATION

| Device       | Package            | Shipping†        |
|--------------|--------------------|------------------|
| P2N2222AG    | TO-92<br>(Pb-Free) | 5000 Units/Bulk  |
| P2N2222ARL1G | TO-92<br>(Pb-Free) | 2000/Tape & Ammo |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

\*For additional information on our Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

# P2N2222A

## ELECTRICAL CHARACTERISTICS ( $T_A = 25^\circ\text{C}$ unless otherwise noted)

| Characteristic                                                                                                                               | Symbol        | Min    | Max        | Unit            |
|----------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|------------|-----------------|
| <b>OFF CHARACTERISTICS</b>                                                                                                                   |               |        |            |                 |
| Collector – Emitter Breakdown Voltage<br>( $I_C = 10\text{ mA}$ , $I_B = 0$ )                                                                | $V_{(BR)CEO}$ | 40     | –          | Vdc             |
| Collector – Base Breakdown Voltage<br>( $I_C = 10\text{ }\mu\text{A}$ , $I_E = 0$ )                                                          | $V_{(BR)CBO}$ | 75     | –          | Vdc             |
| Emitter – Base Breakdown Voltage<br>( $I_E = 10\text{ }\mu\text{A}$ , $I_C = 0$ )                                                            | $V_{(BR)EBO}$ | 6.0    | –          | Vdc             |
| Collector Cutoff Current<br>( $V_{CE} = 60\text{ Vdc}$ , $V_{EB(off)} = 3.0\text{ Vdc}$ )                                                    | $I_{CEX}$     | –      | 10         | nAdc            |
| Collector Cutoff Current<br>( $V_{CB} = 60\text{ Vdc}$ , $I_E = 0$ )<br>( $V_{CB} = 60\text{ Vdc}$ , $I_E = 0$ , $T_A = 150^\circ\text{C}$ ) | $I_{CBO}$     | –<br>– | 0.01<br>10 | $\mu\text{Adc}$ |
| Emitter Cutoff Current<br>( $V_{EB} = 3.0\text{ Vdc}$ , $I_C = 0$ )                                                                          | $I_{EBO}$     | –      | 10         | nAdc            |
| Collector Cutoff Current<br>( $V_{CE} = 10\text{ V}$ )                                                                                       | $I_{CEO}$     | –      | 10         | nAdc            |
| Base Cutoff Current<br>( $V_{CE} = 60\text{ Vdc}$ , $V_{EB(off)} = 3.0\text{ Vdc}$ )                                                         | $I_{BEX}$     | –      | 20         | nAdc            |

## ON CHARACTERISTICS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |                                         |                                   |     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------|-----------------------------------|-----|
| DC Current Gain<br>( $I_C = 0.1\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ )<br>( $I_C = 1.0\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ )<br>( $I_C = 10\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ )<br>( $I_C = 10\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ , $T_A = -55^\circ\text{C}$ )<br>( $I_C = 150\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ ) (Note 1)<br>( $I_C = 150\text{ mA}$ , $V_{CE} = 1.0\text{ Vdc}$ ) (Note 1)<br>( $I_C = 500\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ ) (Note 1) | $h_{FE}$      | 35<br>50<br>75<br>35<br>100<br>50<br>40 | –<br>–<br>–<br>–<br>300<br>–<br>– | –   |
| Collector – Emitter Saturation Voltage (Note 1)<br>( $I_C = 150\text{ mA}$ , $I_B = 15\text{ mA}$ )<br>( $I_C = 500\text{ mA}$ , $I_B = 50\text{ mA}$ )                                                                                                                                                                                                                                                                                                                       | $V_{CE(sat)}$ | –<br>–                                  | 0.3<br>1.0                        | Vdc |
| Base – Emitter Saturation Voltage (Note 1)<br>( $I_C = 150\text{ mA}$ , $I_B = 15\text{ mA}$ )<br>( $I_C = 500\text{ mA}$ , $I_B = 50\text{ mA}$ )                                                                                                                                                                                                                                                                                                                            | $V_{BE(sat)}$ | 0.6<br>–                                | 1.2<br>2.0                        | Vdc |

## SMALL-SIGNAL CHARACTERISTICS

|                                                                                                                                                                                        |           |             |             |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------------|------------------|
| Current – Gain – Bandwidth Product (Note 2)<br>( $I_C = 20\text{ mA}$ , $V_{CE} = 20\text{ Vdc}$ , $f = 100\text{ MHz}$ )C                                                             | $f_T$     | 300         | –           | MHz              |
| Output Capacitance<br>( $V_{CB} = 10\text{ Vdc}$ , $I_E = 0$ , $f = 1.0\text{ MHz}$ )                                                                                                  | $C_{obo}$ | –           | 8.0         | pF               |
| Input Capacitance<br>( $V_{EB} = 0.5\text{ Vdc}$ , $I_C = 0$ , $f = 1.0\text{ MHz}$ )                                                                                                  | $C_{ibo}$ | –           | 25          | pF               |
| Input Impedance<br>( $I_C = 1.0\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ , $f = 1.0\text{ kHz}$ )<br>( $I_C = 10\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ , $f = 1.0\text{ kHz}$ )           | $h_{ie}$  | 2.0<br>0.25 | 8.0<br>1.25 | k $\Omega$       |
| Voltage Feedback Ratio<br>( $I_C = 1.0\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ , $f = 1.0\text{ kHz}$ )<br>( $I_C = 10\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ , $f = 1.0\text{ kHz}$ )    | $h_{re}$  | –<br>–      | 8.0<br>4.0  | $\times 10^{-4}$ |
| Small-Signal Current Gain<br>( $I_C = 1.0\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ , $f = 1.0\text{ kHz}$ )<br>( $I_C = 10\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ , $f = 1.0\text{ kHz}$ ) | $h_{fe}$  | 50<br>75    | 300<br>375  | –                |
| Output Admittance<br>( $I_C = 1.0\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ , $f = 1.0\text{ kHz}$ )<br>( $I_C = 10\text{ mA}$ , $V_{CE} = 10\text{ Vdc}$ , $f = 1.0\text{ kHz}$ )         | $h_{oe}$  | 5.0<br>25   | 35<br>200   | $\mu\text{Mhos}$ |
| Collector Base Time Constant<br>( $I_E = 20\text{ mA}$ , $V_{CB} = 20\text{ Vdc}$ , $f = 31.8\text{ MHz}$ )                                                                            | $rb'C_c$  | –           | 150         | ps               |
| Noise Figure<br>( $I_C = 100\text{ }\mu\text{A}$ , $V_{CE} = 10\text{ Vdc}$ , $R_S = 1.0\text{ k}\Omega$ , $f = 1.0\text{ kHz}$ )                                                      | $N_F$     | –           | 4.0         | dB               |

1. Pulse Test: Pulse Width  $\leq 300\text{ }\mu\text{s}$ , Duty Cycle  $\leq 2.0\%$ .
2.  $f_T$  is defined as the frequency at which  $|h_{fe}|$  extrapolates to unity.

# P2N2222A

## ELECTRICAL CHARACTERISTICS ( $T_A = 25^\circ\text{C}$ unless otherwise noted) (Continued)

| Characteristic                   | Symbol | Min | Max | Unit |
|----------------------------------|--------|-----|-----|------|
| <b>SWITCHING CHARACTERISTICS</b> |        |     |     |      |
| Delay Time                       | $t_d$  | —   | 10  | ns   |
| Rise Time                        | $t_r$  | —   | 25  | ns   |
| Storage Time                     | $t_s$  | —   | 225 | ns   |
| Fall Time                        | $t_f$  | —   | 60  | ns   |

## SWITCHING TIME EQUIVALENT TEST CIRCUITS

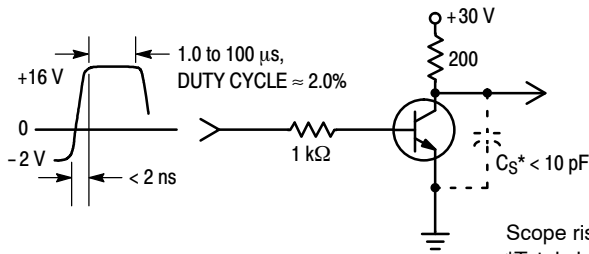


Figure 1. Turn-On Time

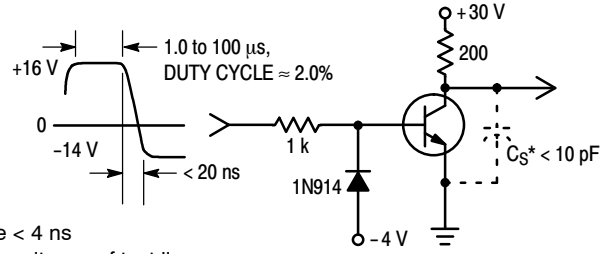


Figure 2. Turn-Off Time

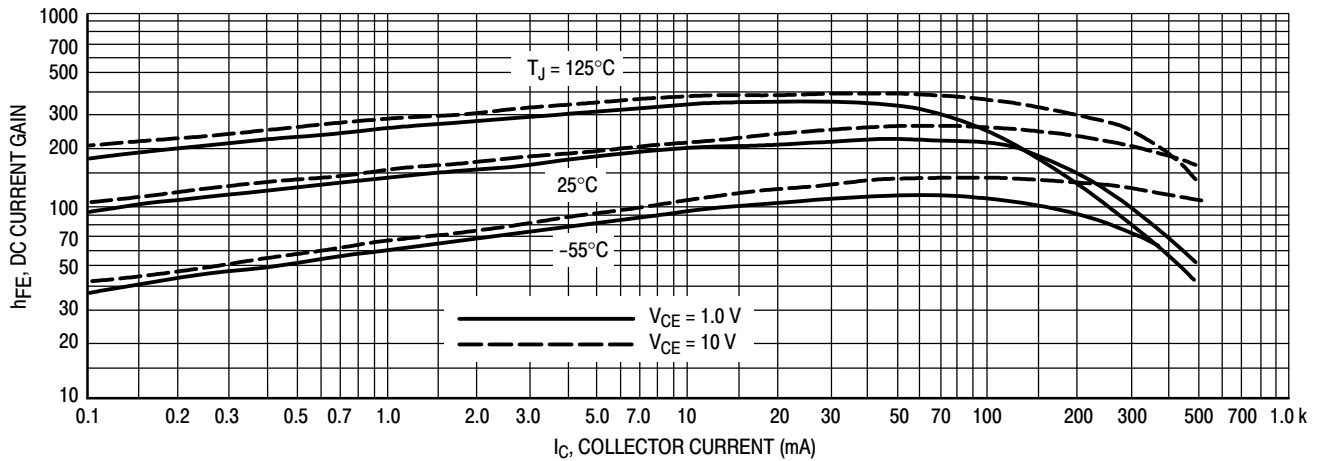
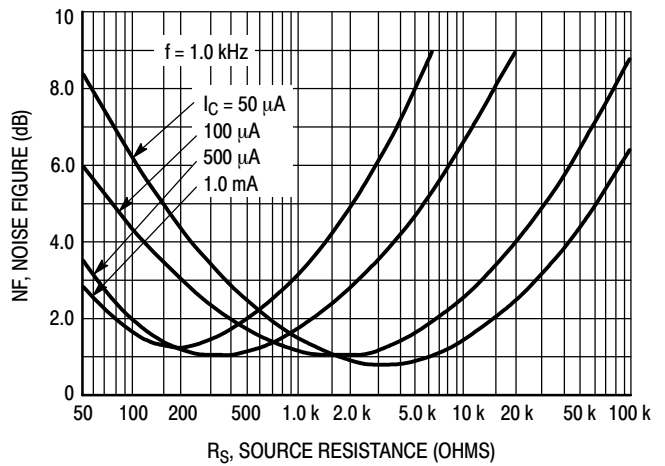
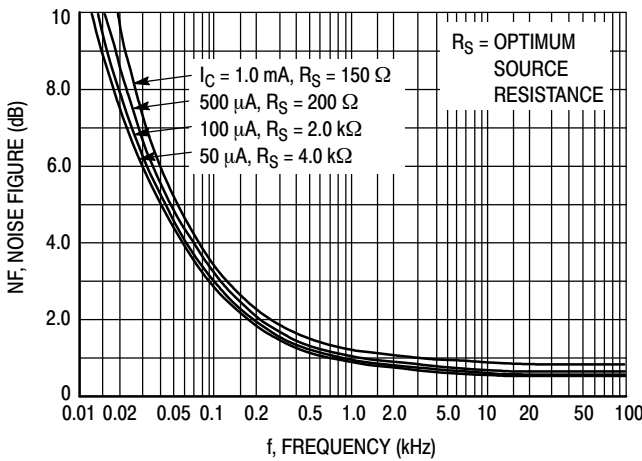
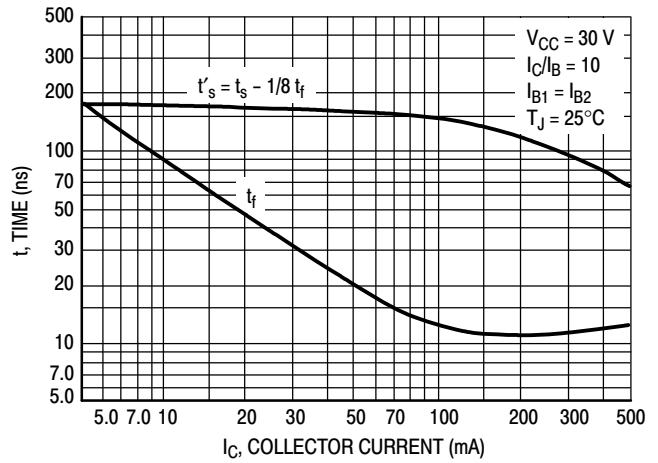
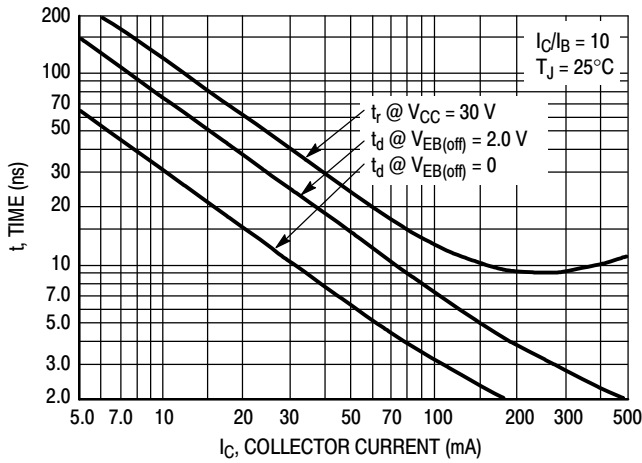
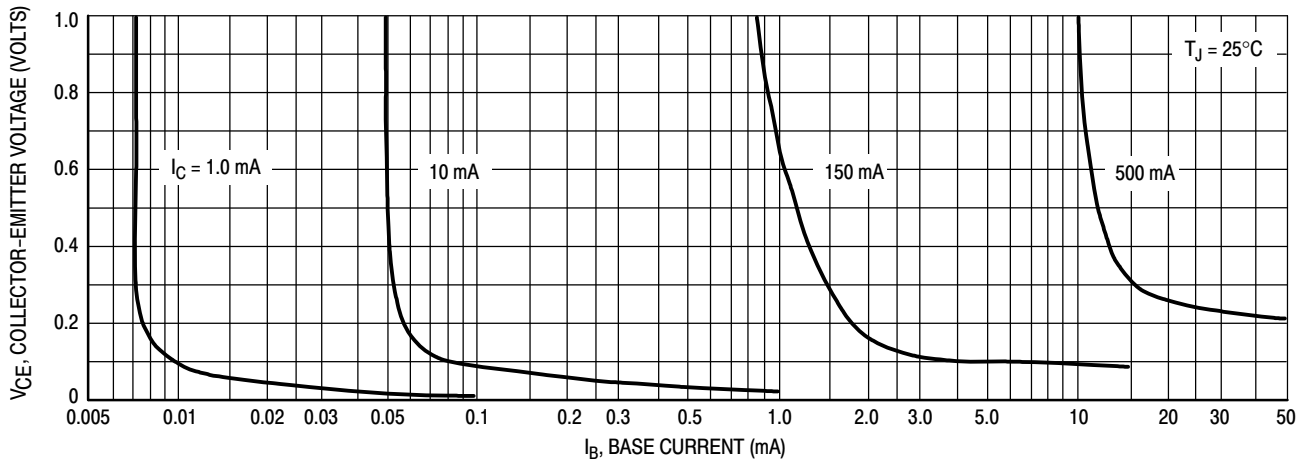


Figure 3. DC Current Gain

# P2N2222A



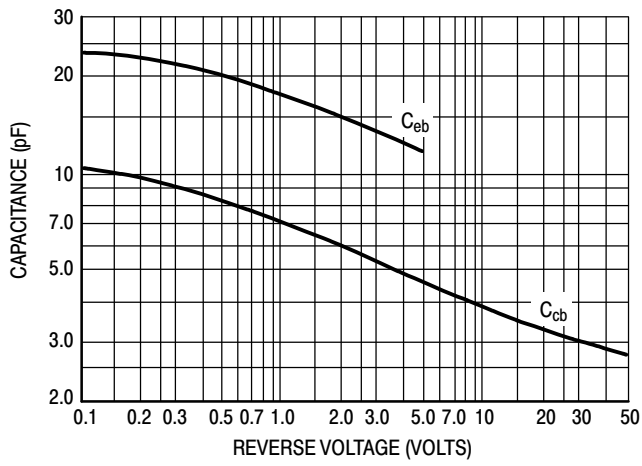


Figure 9. Capacitances

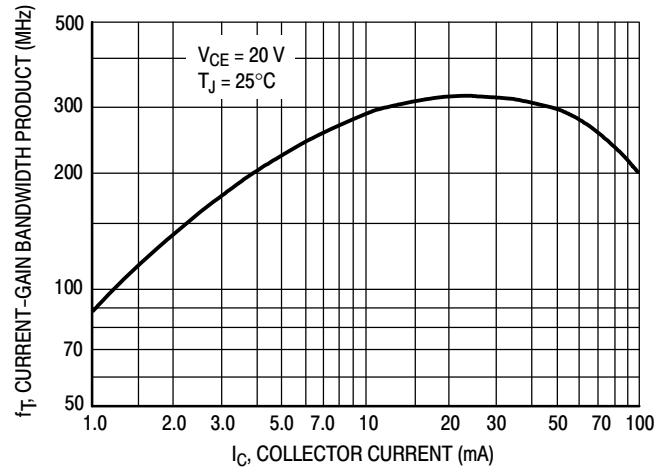


Figure 10. Current-Gain Bandwidth Product

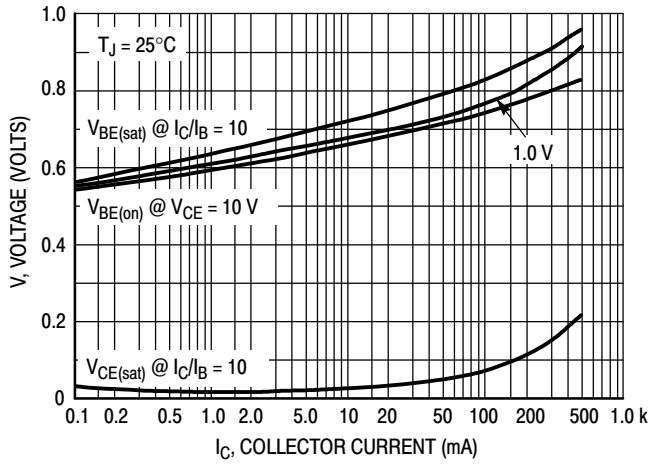


Figure 11. "On" Voltages

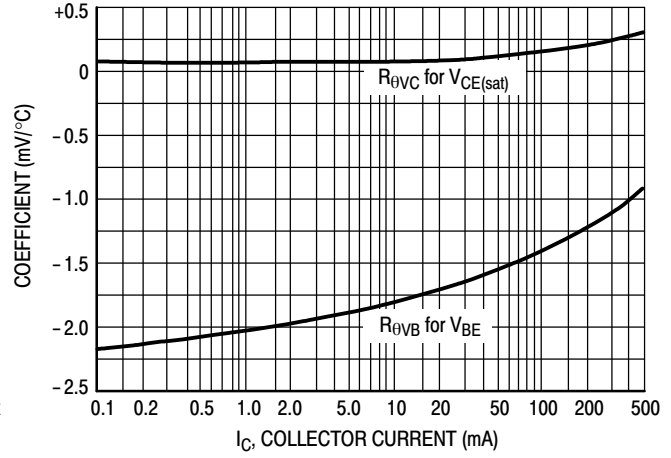
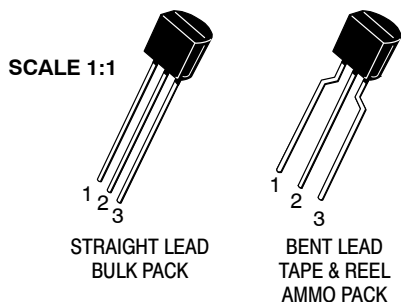
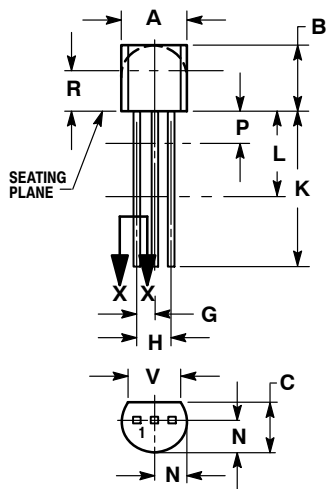


Figure 12. Temperature Coefficients

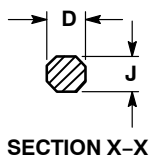


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DATE 09 MAR 2007



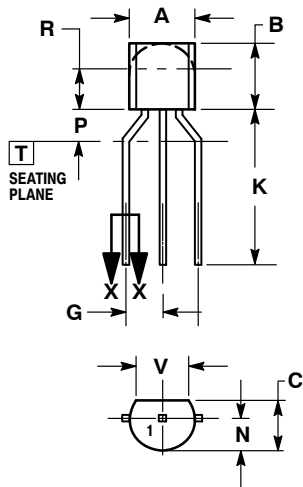
STRAIGHT LEAD  
BULK PACK



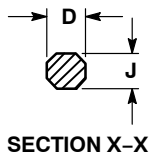
NOTES:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.
3. CONTOUR OF PACKAGE BEYOND DIMENSION R IS UNCONTROLLED.
4. LEAD DIMENSION IS UNCONTROLLED IN P AND BEYOND DIMENSION K MINIMUM.

| DIM | INCHES |       | MILLIMETERS |       |
|-----|--------|-------|-------------|-------|
|     | MIN    | MAX   | MIN         | MAX   |
| A   | 0.175  | 0.205 | 4.45        | 5.20  |
| B   | 0.170  | 0.210 | 4.32        | 5.33  |
| C   | 0.125  | 0.165 | 3.18        | 4.19  |
| D   | 0.016  | 0.021 | 0.407       | 0.533 |
| G   | 0.045  | 0.055 | 1.15        | 1.39  |
| H   | 0.095  | 0.105 | 2.42        | 2.66  |
| J   | 0.015  | 0.020 | 0.39        | 0.50  |
| K   | 0.500  | ---   | 12.70       | ---   |
| L   | 0.250  | ---   | 6.35        | ---   |
| N   | 0.080  | 0.105 | 2.04        | 2.66  |
| P   | ---    | 0.100 | ---         | 2.54  |
| R   | 0.115  | ---   | 2.93        | ---   |
| V   | 0.135  | ---   | 3.43        | ---   |



BENT LEAD  
TAPE & REEL  
AMMO PACK



NOTES:

1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 1994.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. CONTOUR OF PACKAGE BEYOND DIMENSION R IS UNCONTROLLED.
4. LEAD DIMENSION IS UNCONTROLLED IN P AND BEYOND DIMENSION K MINIMUM.

| DIM | MILLIMETERS |      |
|-----|-------------|------|
|     | MIN         | MAX  |
| A   | 4.45        | 5.20 |
| B   | 4.32        | 5.33 |
| C   | 3.18        | 4.19 |
| D   | 0.40        | 0.54 |
| G   | 2.40        | 2.80 |
| J   | 0.39        | 0.50 |
| K   | 12.70       | ---  |
| N   | 2.04        | 2.66 |
| P   | 1.50        | 4.00 |
| R   | 2.93        | ---  |
| V   | 3.43        | ---  |

STYLES ON PAGE 2

|                  |                |                                                                                                                                                                                     |
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| DESCRIPTION:     | TO-92 (TO-226) | PAGE 1 OF 2                                                                                                                                                                         |

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**TO-92 (TO-226)**  
**CASE 29-11**  
**ISSUE AM**

DATE 09 MAR 2007

|                                                                      |                                                                             |                                                                     |                                                                        |                                                                    |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>STYLE 1:</b><br>PIN 1. EMITTER<br>2. BASE<br>3. COLLECTOR         | <b>STYLE 2:</b><br>PIN 1. BASE<br>2. EMITTER<br>3. COLLECTOR                | <b>STYLE 3:</b><br>PIN 1. ANODE<br>2. ANODE<br>3. CATHODE           | <b>STYLE 4:</b><br>PIN 1. CATHODE<br>2. CATHODE<br>3. ANODE            | <b>STYLE 5:</b><br>PIN 1. DRAIN<br>2. SOURCE<br>3. GATE            |
| <b>STYLE 6:</b><br>PIN 1. GATE<br>2. SOURCE & SUBSTRATE<br>3. DRAIN  | <b>STYLE 7:</b><br>PIN 1. SOURCE<br>2. DRAIN<br>3. GATE                     | <b>STYLE 8:</b><br>PIN 1. DRAIN<br>2. GATE<br>3. SOURCE & SUBSTRATE | <b>STYLE 9:</b><br>PIN 1. BASE 1<br>2. EMITTER<br>3. BASE 2            | <b>STYLE 10:</b><br>PIN 1. CATHODE<br>2. GATE<br>3. ANODE          |
| <b>STYLE 11:</b><br>PIN 1. ANODE<br>2. CATHODE & ANODE<br>3. CATHODE | <b>STYLE 12:</b><br>PIN 1. MAIN TERMINAL 1<br>2. GATE<br>3. MAIN TERMINAL 2 | <b>STYLE 13:</b><br>PIN 1. ANODE 1<br>2. GATE<br>3. CATHODE 2       | <b>STYLE 14:</b><br>PIN 1. EMITTER<br>2. COLLECTOR<br>3. BASE          | <b>STYLE 15:</b><br>PIN 1. ANODE 1<br>2. CATHODE<br>3. ANODE 2     |
| <b>STYLE 16:</b><br>PIN 1. ANODE<br>2. GATE<br>3. CATHODE            | <b>STYLE 17:</b><br>PIN 1. COLLECTOR<br>2. BASE<br>3. EMITTER               | <b>STYLE 18:</b><br>PIN 1. ANODE<br>2. CATHODE<br>3. NOT CONNECTED  | <b>STYLE 19:</b><br>PIN 1. GATE<br>2. ANODE<br>3. CATHODE              | <b>STYLE 20:</b><br>PIN 1. NOT CONNECTED<br>2. CATHODE<br>3. ANODE |
| <b>STYLE 21:</b><br>PIN 1. COLLECTOR<br>2. EMITTER<br>3. BASE        | <b>STYLE 22:</b><br>PIN 1. SOURCE<br>2. GATE<br>3. DRAIN                    | <b>STYLE 23:</b><br>PIN 1. GATE<br>2. SOURCE<br>3. DRAIN            | <b>STYLE 24:</b><br>PIN 1. EMITTER<br>2. COLLECTOR/ANODE<br>3. CATHODE | <b>STYLE 25:</b><br>PIN 1. MT 1<br>2. GATE<br>3. MT 2              |
| <b>STYLE 26:</b><br>PIN 1. $V_{CC}$<br>2. GROUND 2<br>3. OUTPUT      | <b>STYLE 27:</b><br>PIN 1. MT<br>2. SUBSTRATE<br>3. MT                      | <b>STYLE 28:</b><br>PIN 1. CATHODE<br>2. ANODE<br>3. GATE           | <b>STYLE 29:</b><br>PIN 1. NOT CONNECTED<br>2. ANODE<br>3. CATHODE     | <b>STYLE 30:</b><br>PIN 1. DRAIN<br>2. GATE<br>3. SOURCE           |
| <b>STYLE 31:</b><br>PIN 1. GATE<br>2. DRAIN<br>3. SOURCE             | <b>STYLE 32:</b><br>PIN 1. BASE<br>2. COLLECTOR<br>3. EMITTER               | <b>STYLE 33:</b><br>PIN 1. RETURN<br>2. INPUT<br>3. OUTPUT          | <b>STYLE 34:</b><br>PIN 1. INPUT<br>2. GROUND<br>3. LOGIC              | <b>STYLE 35:</b><br>PIN 1. GATE<br>2. COLLECTOR<br>3. EMITTER      |

|                         |                       |                                                                                                                                                                                     |
|-------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| <b>DESCRIPTION:</b>     | <b>TO-92 (TO-226)</b> | <b>PAGE 2 OF 2</b>                                                                                                                                                                  |

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